Project 2: Content-based Image Retrieval

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The project is about Content based image retrieval, given a database of images and a target image, find images in the data with similar content without using machine learning methods. Instead, we will focus on more generic characteristics of the images such as color, texture, and their spatial layout. This runs in two phases, the first phase is where we read all the images the given directory and extract the feature vectors for every image and stores in a csv file. The second phase is to retrieve every feature vector to the feature set and then outputs the top N matches.

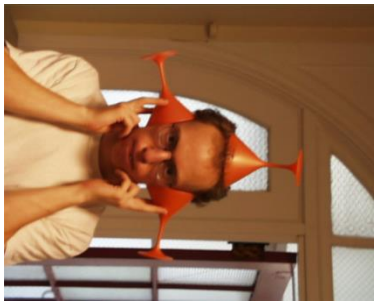
Task-1: Baseline Matching

In baseline matching, the feature vector is made with the 9X9 pixels in the middle of the image. The program requires the target image, feature set and the feature vector file then it outputs the top 3 images. The feature vector file is iterated and compared to the feature set using distance metric which is the sum-of-squared-difference, then outputs the top 3 images of least difference.

The top 3 images for the target image pic.1016.jpg image are:

Target image:

Top 3 matches:



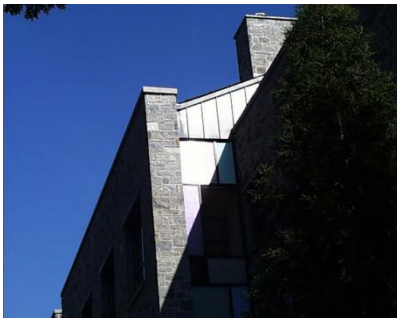
Task 2: Histogram Matching

In the histogram matching task, the feature vector is made by constructing the 2D histogram for the whole image. The program requires the target image, feature set and the feature vector file then it outputs the top 3 images. The feature vector file is iterated and compared to the feature set using distance metric which is the histogram interaction, then outputs the top 3 images with highest intersection.

The top 3 images for the target image pic.0164.jpg image are:

Target image:

Output images:



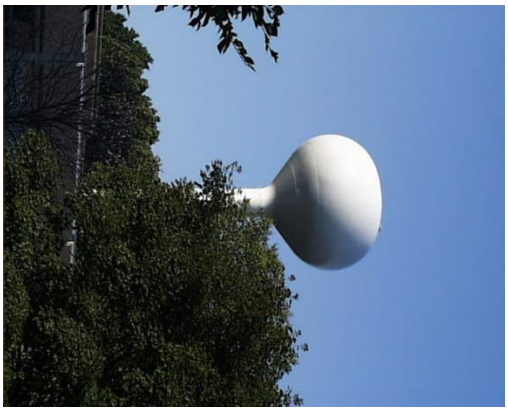
Task 3: Multi-histogram Matching

In the multi-histogram matching task, the feature vector is made by constructing two 2D histograms, one for the whole image and for the region of interest in the image. The program requires the target image, feature set and the feature vector file then it outputs the top 3 images. The feature vector file is iterated and compared to the feature set using distance metric which is the histogram interaction, then outputs the top 3 images with highest intersection.

The top 3 images for the target image pic.0274.jpg image are:

Target image:

Output images:



Task 4: Texture and Color

In the texture and color matching task, the feature vector is made by constructing two histograms, is the color histogram and the other one is after applying the Sobel filters to the images and using the magnitude and orientation. The program requires the target image, feature set and the feature vector file then it outputs the top 3 images. The feature vector file is iterated and compared to the feature set using distance metric which is the histogram interaction, then outputs the top 3 images with highest intersection.

The top 3 images for the target image pic.0535.jpg image are:

Target image:

Output images:

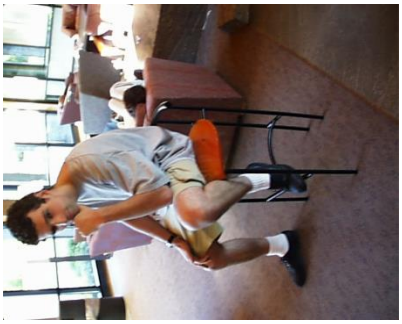
Task 5: Custom Design

In the custom design task, the user inputs two images containing the same object and the program tries to find the similar objects. In this program the feature vector is made by taking average of the color histograms and texture histograms of both images. The feature vector file is iterated and compared to the feature set using distance metric which is the histogram interaction, then outputs the top 10 images with highest intersection.

The top 10 images for the target images pic.0343.jpg and pic.0345.jpg image are:

Target images:

Output images:

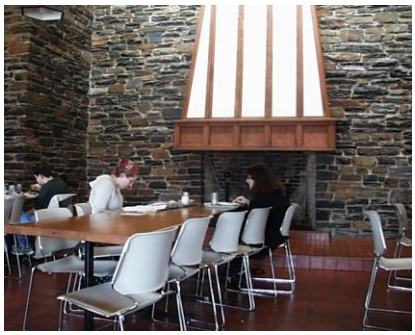


Extension work:

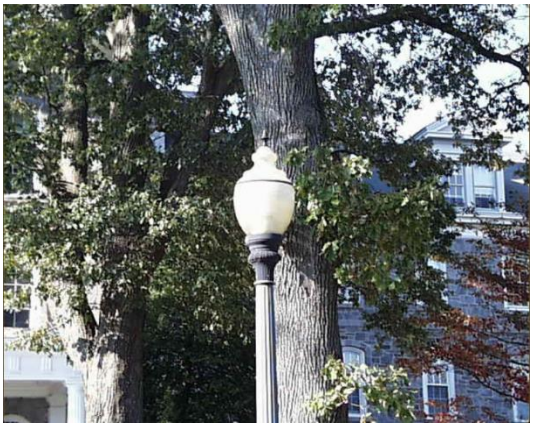
As the extension, I applied L5E5 filter on the image and created a histogram using the magnitude and orientation and the other histogram used is color histogram. The program requires the target image, feature set and the feature vector file then it outputs the top 3 images. The feature vector file is iterated and compared to the feature set using distance metric which is the histogram interaction, then outputs the top 3 images with highest intersection.

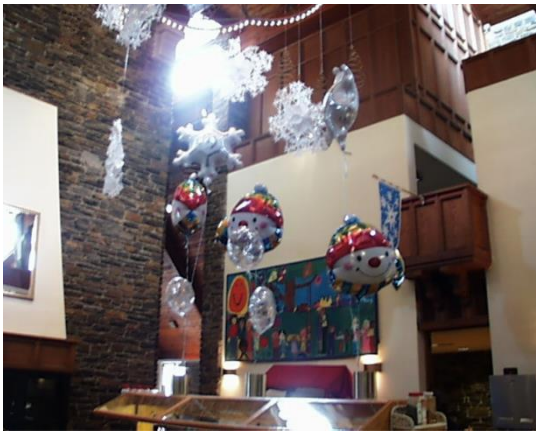
The top 3 images for the target image pic.0535.jpg image are:

Target image:



Output images:





Reflections and Acknowledgement:

From this project, I learned How to access the images from a directory and store the feature vectors into a .csv file. I also learned how to compare images by plotting color histograms and texture histograms. As I wrote most of the code without using inbuilt functions it is very useful to get deeper knowledge on the content based image retrieval.

When working on the assignment, I discussed with the TA on how to plot texture histograms and I got cleared.